

ICT702 Data Visualisation

Telling the Truth with Data Visualisation

Module 9





I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



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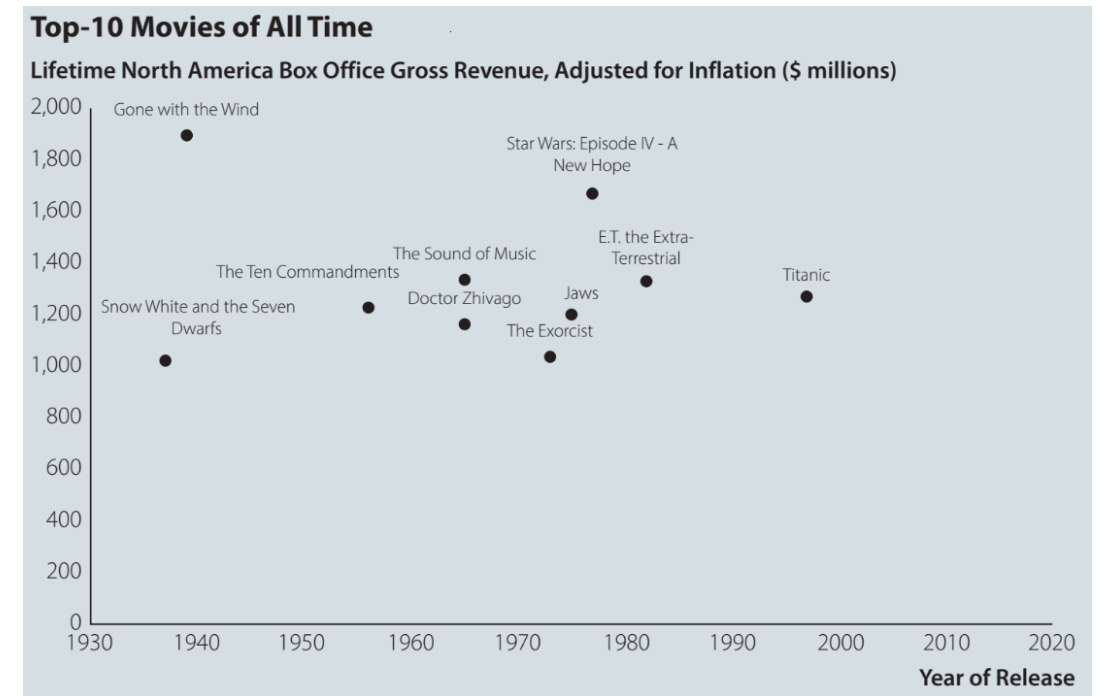
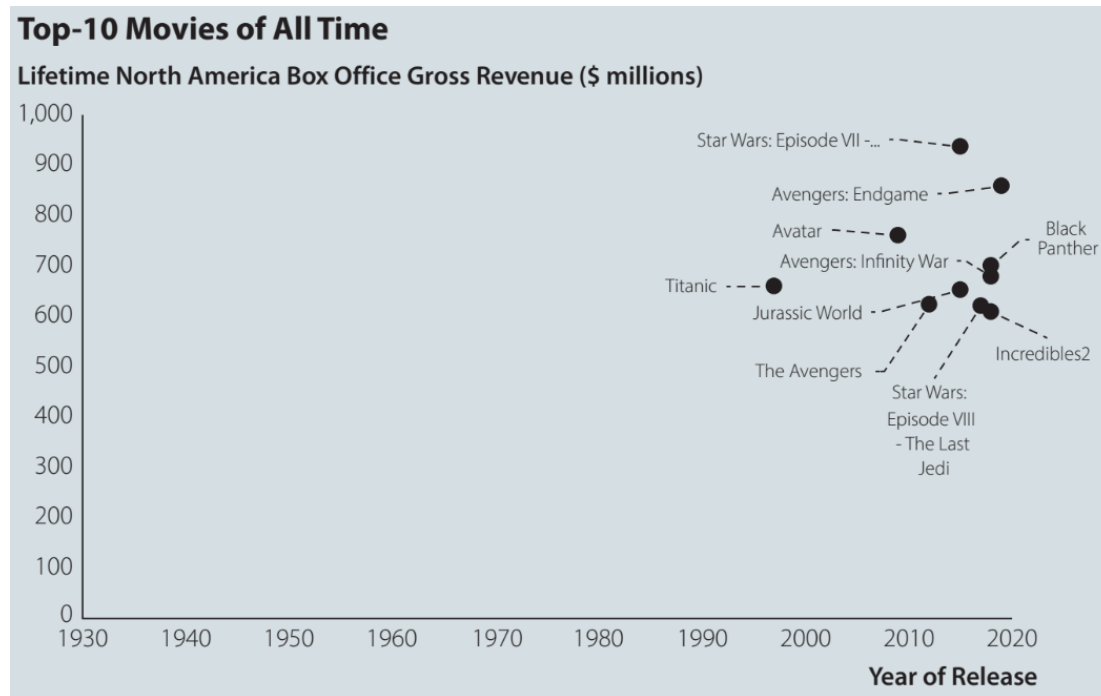
Learning Objectives for Module 9

- Identify missing data and data errors using Excel.
- Define the meaning of biased data and explain the concepts of selection bias and survivor bias.
- Define Simpson's paradox and explain how a scatter chart can be used to identify some instances of Simpson's paradox.
- Explain the importance of adjusting for inflation in time series data that represent long time periods and use a price index to adjust nominal values to account for inflation.
- Identify deceptive design practices related to the axes used in charts and suggest ways to improve the axes in these charts to communicate insights to the audience more clearly.
- Explain why dual-axis charts are often confusing and misleading for audiences and suggest an alternative to using a dual-axis chart that is less confusing for the audience.
- Explain how the range of data and the temporal frequency of data included in a chart affects the insights conveyed to the audience.
- Explain why some geographic maps can result in misleading data visualisations and provide recommendations for how to improve these types of maps.

Part 1

Data Visualisation Makeover

Scatter charts showing effect of inflation adjustment on gross revenues for lifetime North America box office top-10 movies of all time



Source: https://help.imdb.com/article/imdbpro/industry-research/box-office-mojo-by-imdbpro-faq/GCWTV4MQKGWRAUAP?ref_=mojo_cso_md#inflation.

Missing Data and Data Errors

A portion of an Excel spreadsheet showing Blakely Tires data

	A	B	C	D	E
1	ID Number	Position on Automobile	Life of Tire (Months)	Tread Depth	Miles
2	13391487	LR	58.4	2.2	2805
3	21678308	LR	17.3	8.3	39371
4	18414311	RR	16.5	8.6	13367
5	19778103	RR	8.2	9.8	1931
6	16355454	RR	13.7	8.9	23992
7	8952817	LR	52.8	3	48961
8	6559652	RR	14.7	8.8	4585
9	16289814	LR	6.2	10.1	5221

Company: Blakely Tires is a producer of vehicle tires located in the United States.

Study objective: to learn about the conditions of its tires on vehicles in the state of Texas.

Issue: assess data quality.

Identifying Missing Data by Counting Data Blanks

Portion of Excel spreadsheet showing number of missing values for variables in Blakely Tires data

	A	B	C	D	E	F	G	H	I	J
1	ID Number	Position on Automobile	Life of Tire (Months)	Tread Depth	Miles			Life of Tire (Months)	Tread Depth	Miles
2	13391487	LR	58.4	2.2	2805		Number Missing Values	=COUNTBLANK(C2:C457)	=COUNTBLANK(D2:D457)	=COUNTBLANK(E2:E457)
3	21678308	LR	17.3	8.3	39371					
4	18414311	RR	16.5	8.6	13367					
5	19778103	RR	8.2	9.8	1991					
6	16355454	RR	13.7	8.9	23992					
7	8952817	LR	52.8	3	48961					
8	6559652	RR	14.7	8.8	4585					
9	16289814	LR	6.2	10.1	5221					

	A	B	C	D	E	F	G	H	I	J
1	ID Number	Position on Automobile	Life of Tire (Months)	Tread Depth	Miles			Life of Tire (Months)	Tread Depth	Miles
2	13391487	LR	58.4	2.2	2805		Number Missing Values	0	0	1
3	21678308	LR	17.3	8.3	39371					
4	18414311	RR	16.5	8.6	13367					
5	19778103	RR	8.2	9.8	1991					
6	16355454	RR	13.7	8.9	23992					
7	8952817	LR	52.8	3.0	48961					
8	6559652	RR	14.7	8.8	4585					
9	16289814	LR	6.2	10.1	5221					

To count the missing observations for each variable in the file *BlakelyTires*, use the formula

=COUNTBLANK(C2:C457)
in cell H2, and copy cell H2 to cells I2 and J2

Identifying Missing Data Using Conditional Formatting

Portion of Excel spreadsheet showing Blakely Tires data sorted from lowest to highest by ID number and missing value highlighted

	A	B	C	D	E
52	2253516	LF	26.2	7.1	55231
53	2253516	RR	26.2	7.1	55231
54	3121851	LR	17.1	8.4	21378
55	3121851	RR	17.1	8.4	21378
56	3121851	RF	17.1	8.4	21378
57	3121851	LF	17.1	8.5	21378
58	3354942	LF	17.1	8.5	
59	3354942	RF	21.4	7.7	33254
60	3354942	RR	21.4	7.8	33254
61	3354942	LR	21.4	7.7	33254
62	3574739	RR	73.3	0.2	57313

To highlight the missing observations for each variable in the *BlakelyTires* data file, use the Excel Conditional Formatting tool.

Identifying Extreme Data Errors

Portion of Excel spreadsheet showing Blakely Tires data sorted on Life of Tires (Months) and calculated summary statistics

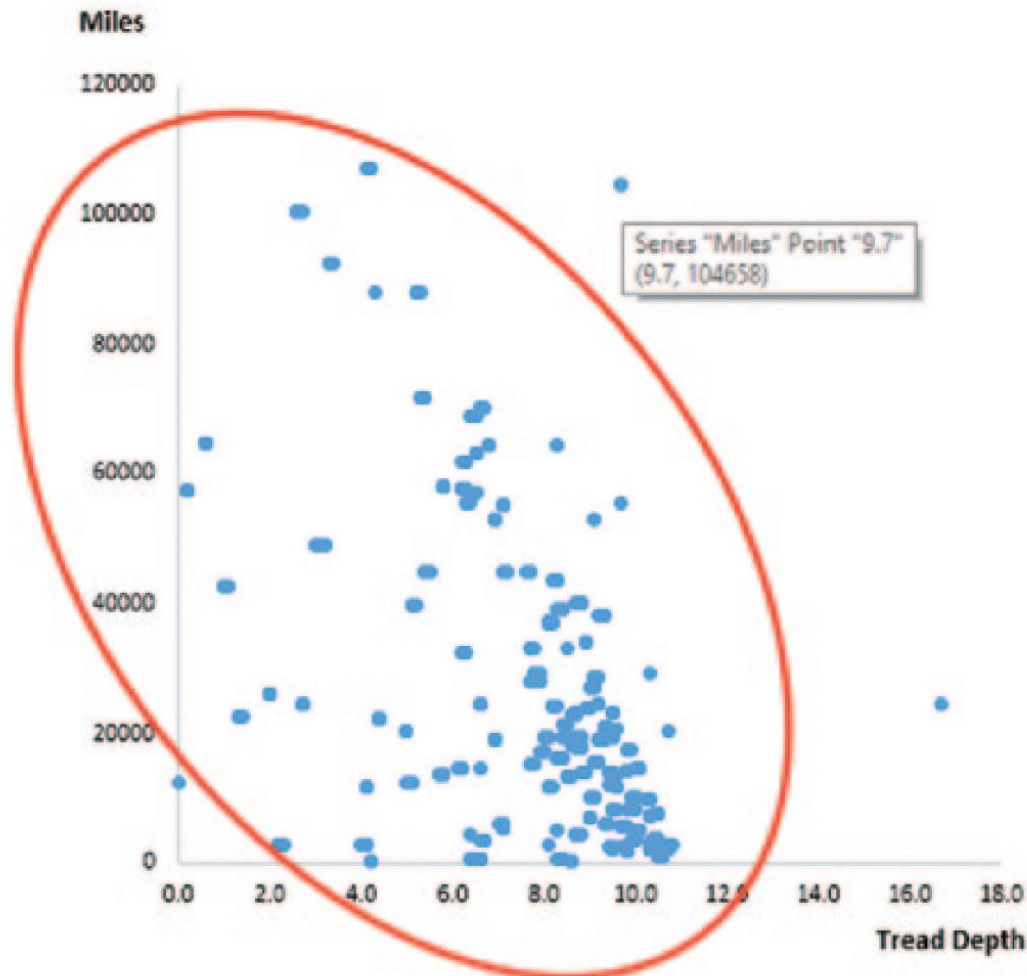
	A	B	C	D	E	F	G	H	I	J
1	ID Number	Position on Automobile	Life of Tire (Months)	Tread Depth	Miles			Life of Tire (Months)	Tread Depth	Miles
2	9091771	RR	1.8	10.7	2917		Number Missing Values	0	0	0
3	9091771	LF	1.8	10.7	2917		Mean	29.76	7.64	25834.54
4	9091771	RF	1.8	10.8	2917		Standard Deviation	31.83	2.51	24143.38
5	7712178	LR	2.1	10.6	2186		Minimum	1.8	0.0	206.0
6	7712178	LF	2.1	10.7	2186		Maximum	601.0	16.7	107237.0
7	7712178	RR	2.1	10.7	2186					
452	3574739	RR	73.3	0.2	57313					
453	3574739	RF	73.3	0.2	57313					
454	3574739	LF	73.3	0.2	57313					
455	3574739	LR	73.3	0.2	57313					
456	2122934	LR	111.0	9.3	21000					
457	8696859	LR	601.0	2.0	26129					

Use summary statistics to detect suspicious values.

- *Life of Tire*: a maximum value of 601 months for a left rear tire. Comparison with the other tires suggests a value of 60.1 months instead.
- *Tread Depth*: a minimum value of 0.0/12ths (undrivable) and a maximum value of 16.7/12ths, which exceeds tread depth on new tires.

Identifying Non-Extreme Data Errors

Scatter chart for the Blakely Tires data used to identify possible data errors



Not all erroneous values in a data set are extreme.

To identify non-extreme erroneous values, investigate relationships between quantitative variables using data visualisation tools such as scatter charts.

End of Part 1